Precision Career Coaching: Fine-Tuning Large Language Models for Strategic Job Matching

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Description of the Idea

The core idea of this project is to develop a specialized Generative AI Career Coach that goes beyond simple keyword matching to provide deep, contextual alignment between a candidate's resume and a specific job description. Current general-purpose AI models often struggle with the "nuance" of professional experience; they may fail to recognize how a specific technical skill in one industry translates to another, or they may provide generic advice that does not help a resume pass through modern Applicant Tracking Systems (ATS).

This application will leverage a fine-tuned GAI model trained on a curated dataset of high-performing resumes and corresponding job descriptions. The goal is to create an agent that can analyze a job description, identify the "hidden" requirements that a human recruiter looks for, and then provide a line-by-line optimization of a user's resume. The final product will be a Streamlit-based web application where users can upload their resume and a target job link to receive a "Match Score," a keyword gap analysis, and a set of rewritten bullet points that use the STAR (Situation, Task, Action, Result) method to highlight relevant experience.

Project Goals

The primary goal of this project is to transition a generative model from providing a general solution into a specialized career coaching expert. While base models are capable of basic text manipulation, they often lack the deep context required to understand industry specific jargon and evolving hiring trends. By providing the model with a specialized knowledge base, I aim to teach it how to distinguish between filler text and high-impact professional accomplishments. This encoding process will ensure the AI understands the subtle differences in expectations between various sectors, such as the specific terminology preferred in Data Science versus the leadership language expected in corporate management.

A second technical goal is to successfully perform a full fine-tuning of a GenAI model. A custom dataset which will consist of carefully curated resume and job description pairs that illustrate the ideal alignment between a candidate and a role will be used for this purpose. This will be done in python and I will manage the end-to-end training process, which includes data cleaning, formatting the structural files, and monitoring the training epochs to ensure the model converges on a high-quality solution without over-fitting.

Furthermore, I am setting a goal for measurable optimization to ensure the project provides tangible value. The fine-tuned model should be able to identify at least 90% of the critical keywords and core competencies found within a provided job description. Beyond simple identification, the model must learn to suggest naturally integrated placements for these terms within the existing resume. It is really important that these optimizations do not result in keyword stuffing or the loss of the authentic professional voice that resume had in it. The main goal is to produce a better version of the resume that remains effective and readable to human recruiters while passing through automated filters in ATS.

Finally, I plan to build and deploy a Stream-lit web interface to make this advanced AI accessible to users. This interface will transform the fine-tuned model into a practical tool for job seekers through a seamless upload-and-analyze workflow. This deployment phase will involve integrating the GenAI API into a Python-based web framework to ensure the application is responsive and intuitive. By surfacing the model in a professional web application, I can demonstrate the full lifecycle of a generative AI product from initial data preparation to a finished user-facing tool.

Initial Approach to the Problem

My initial approach for this project starts with a focused phase of data collection and curation. I realized early in the planning process that a generative model is only as good as the examples it learns from, so I intend to build a high-quality dataset of instruction-response pairs. This will involve gathering various professional resumes and their corresponding job descriptions to create a gold standard for what a perfect match looks like. This stage is crucial because I want the AI to learn the art of resume writing rather than just the basic mechanics.

Once the data is ready, I will move into the technical core of the project by performing a full fine-tuning of a GenAI model. During this process, I will be keeping a close eye on the training logs to make sure things are progressing as expected. It is important to me that I learn how to spot signs of over-fitting early because I want the model to be flexible enough to handle resumes from different industries rather than just memorizing a few specific examples. This phase is where I expect to do the most learning by doing as I figure out the best balance between training iterations and data variety.

The final part of my approach involves bringing this model to life through a Streamlit web application. I want to create a clean and intuitive interface where a user can simply upload a file or paste their resume text alongside a job link. My vision for the app is to provide a side-by-side comparison that highlights exactly where the gaps are and how the AI suggested closing them. I might be too optimistic, and I am aware that my approach will shift as I will start working on the project, but I want to share what I want to build. I am excited to see how a model I trained myself can help me and a job seeker to navigate the complexities of the modern job market.

References